SAGAR SACHDEV

in/sagar-sachdev | 925-303-7030 | sagar22sachdev@gmail.com | sagars2.github.io

WORK EXPERIENCE

FieldAI February 2026 – Present
Forward Deployed Robotics Engineer Irvine, CA

- Led forward deployment for 2 largest accounts: **9 sites, 20+ Boston Dynamics Spot robots**
- Ported PTZ tracking node from Python to C++: **3.5x less CPU**, 2.9x less memory, **14x faster startup**
- Migrated production robots from ROS1 to ROS2 on site, preserving ~25 hrs of operator data
- Root-caused localization drift, CPU, and network faults; drove fleet-wide advisories and hotfixes
- Owned **8 release lines**; built release-validation framework adopted by engineering and QA
- Closed field-to-product loop: ran RCA with autonomy teams and fed findings into product proposals

Path Robotics September 2023 – February 2026
Robotics Deployment Engineer Columbus, OH

- Wrote utilities for calibrations, **reducing autonomous cell welding downtime by 50%**, resulting in efficient and streamlined process using **Bash, Python and ROS**
- Created and maintained configuration files for autonomous welding cells, optimizing CAD data using **Blender** and exporting to **Collada** format for **URDF** generation
- Developed **C++ plugin** that converts **Denavit-Hartenberg (DH) parameters** into **XYZRPY** representations in under **20 ms per joint**, achieving **80% reduction in processing time** compared to baseline
- Combined **MetaAI's SAM algorithm** with classical computer vision techniques to segment cell components and bridge the sim-to-real gap; **Improved URDF validation times by 40%**

RESEARCH EXPERIENCE

The AirLab at CMU August 2022 – May 2023
Graduate Research Assistant Pittsburgh, PA

- Developed and validated **time-optimal paths** for a curvature constrained dynamics model in uniform wind conditions for fixed-wing UAVs, with over **37% speed improvement** in path solving speeds using **C++**
- Executed a decision matrix and closed-form solutions from published research for feasible trajectories in zero-wind conditions, achieving a **10x increase in path solving speed**

Biorobotics Lab at CMU August 2021 – August 2022
Graduate Research Assistant Pittsburgh, PA

- Created **software suite for autonomous disassembly** station, **improving fastener removal accuracy to 95%** using **ROS, Python and C++**
- Streamlined **forward kinematics** calculations for a **6-DOF General Stewart Manipulator** in **MATLAB and Python** by utilizing an offline lookup table, leading to 25% reduction in runtime

EDUCATION

Carnegie Mellon University Pittsburgh, PA
Master of Science in Mechanical Engineering Conferred: May 2023
Concentration: Robotics

University of Illinois at Urbana-Champaign Champaign, IL
Bachelor of Science in Engineering Mechanics Conferred: May 2021
Concentration: Mechatronics and Design

PUBLICATIONS

Brady Moon*, **Sagar Sachdev***, Junbin Yuan, Sebastian Scherer, "**Time-Optimal Path Planning in a Constant Wind for Uncrewed Aerial Vehicles using Dubins Set Classification**," *IEEE Robotics and Automation Letters, 2023.* arXiv: 2306.11845v2.

SKILLS

Programming Languages: Python, C/C++, MATLAB/Simulink, HTML/CSS, LaTeX, Bash, Zsh

Software: Linux, Github, ROS, FoxGlove, Docker, Jupyter, Pandas, Scipy, OpenCV, LightGlue, CloudCompare, Jira